**Case Study 06: Tiered Electricity Billing System**

## 1. Problem Statement

Develop a Python program to calculate the monthly electricity charges for a domestic consumer based on a graduated slab tariff system. The program prompts the user to enter the total units of electricity consumed, verifies that the input reading is non-negative, computes tiered energy charges across consumption slabs, adds a fixed meter service charge of Rs. 50, levies an additional 5% surcharge if total consumption exceeds 300 units, and outputs an itemized bill breakdown.

## 2. Objectives

- To validate consumer consumption data against negative numerical inputs.
- To implement multi-tier progressive pricing algorithms using conditional ladder logic (if-elif-else).
- To account for fixed service fees and conditionally apply percentage-based surcharges on high-volume consumption.
- To compute, format, and display the breakdown between base energy charges and the final payable bill.

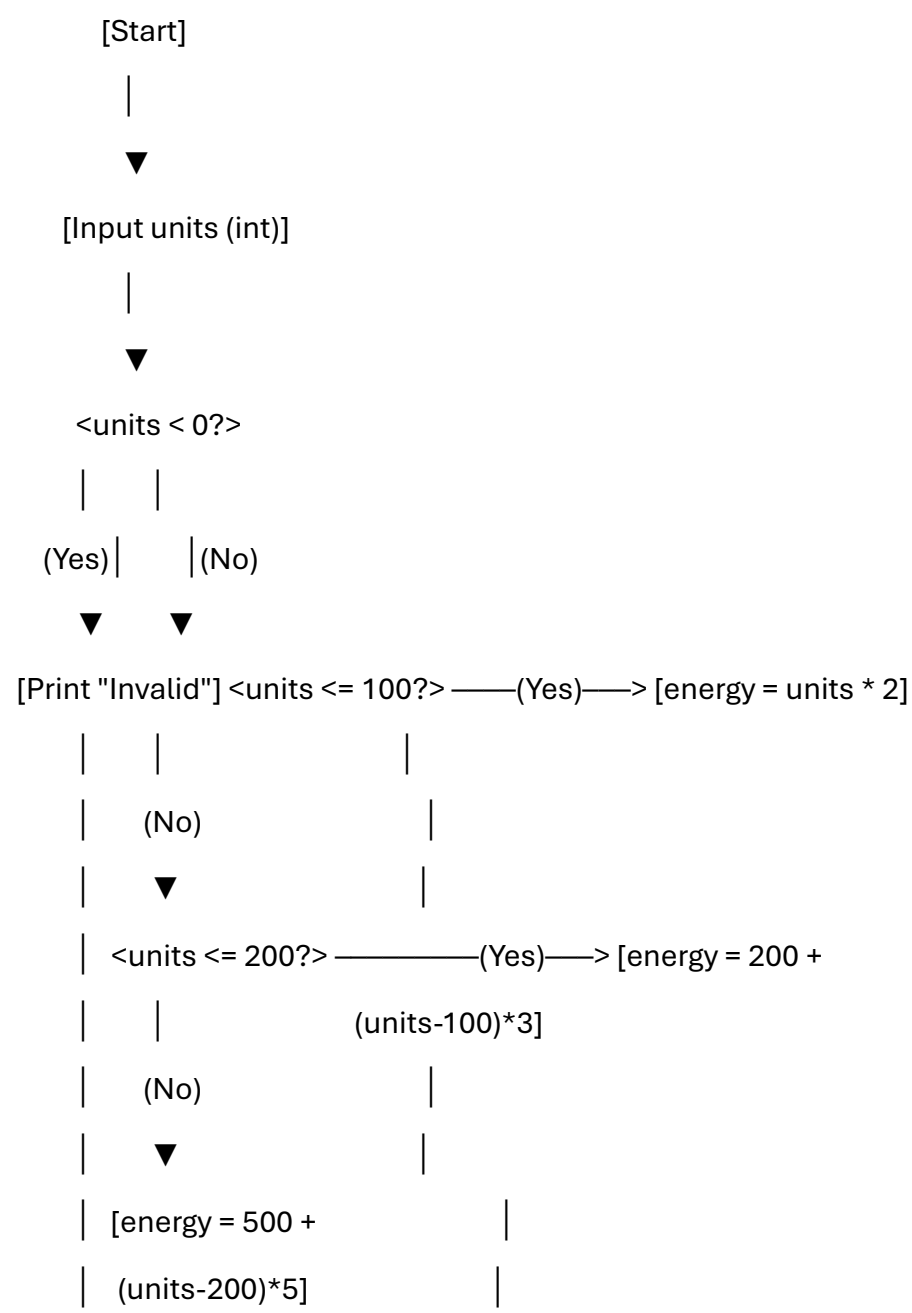
**3. Assumptions**

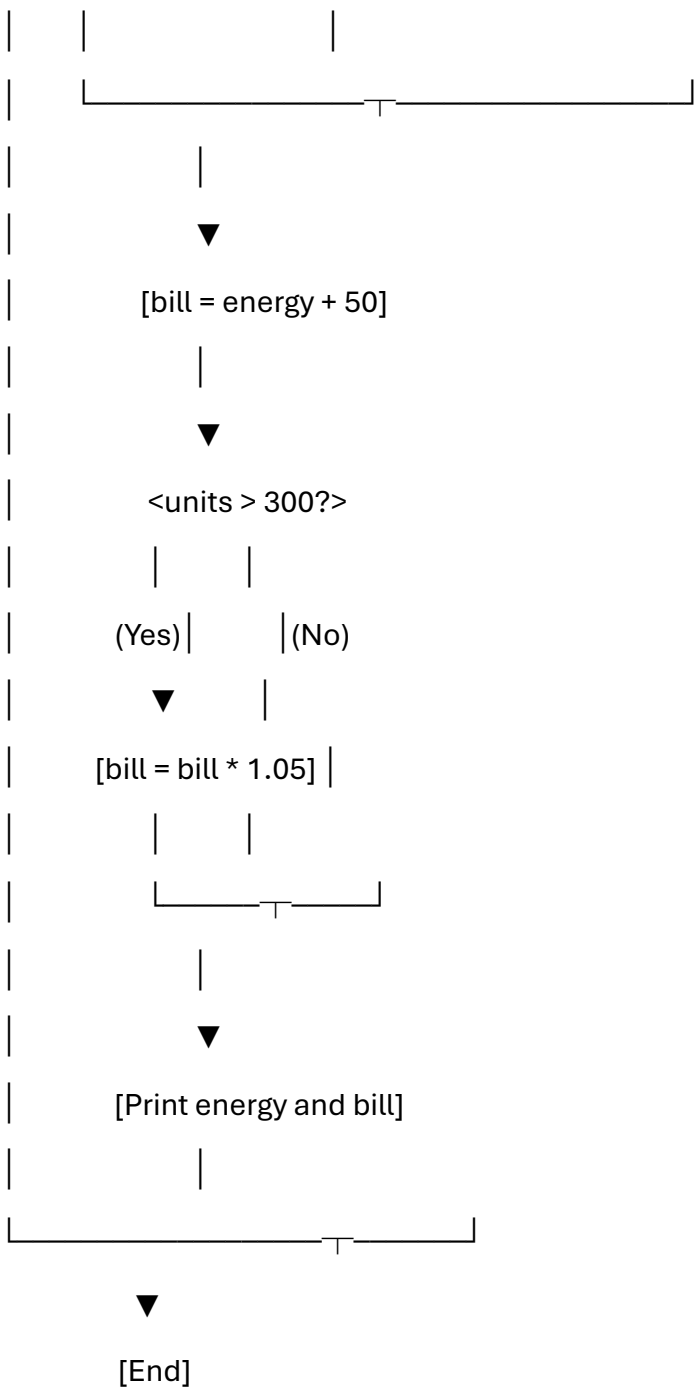
- Units consumed are recorded as integer values.
- The progressive tariff structure applies as follows:
    - First 100 units: Rs. 2.00 per unit.
    - Next 100 units (101 to 200): Rs. 3.00 per unit.
    - Beyond 200 units: Rs. 5.00 per unit.
- A flat fixed surcharge of Rs. 50 is added to every valid bill.
- If consumption strictly exceeds 300 units, a 5% surcharge is levied on the entire aggregate sum (energy charges + fixed fee).
- All final currency amounts are rounded to two decimal places.

**4. Algorithm**

- **Step 1:** Start the program.
- **Step 2:** Prompt the user to input electricity consumption: units = int(input(...)).
- **Step 3:** If units < 0, print "Invalid units" and proceed to Step 9.
- **Step 4:** Calculate energy charges based on consumption slabs:
    - If units <= 100: energy = units * 2.
    - Else if units <= 200: energy = 100 * 2 + (units - 100) * 3.
    - Else: energy = 100 * 2 + 100 * 3 + (units - 200) * 5.
- **Step 5:** Add fixed meter service charge: bill = energy + 50.
- **Step 6:** Check if units > 300. If true, apply surcharge: bill = bill * 1.05.
- **Step 7:** Display the computed energy charge: "Energy charge: Rs. <round(energy, 2)>".
- **Step 8:** Display the total payable amount: "Total bill: Rs. <round(bill, 2)>".
- **Step 9:** Stop the program.

**5. Flowchart Logic / Diagram**






**6. Pseudocode**

```
Plaintext
BEGIN
  PRINT "Enter units consumed: "
  READ units

  IF units < 0 THEN
    PRINT "Invalid units"
  ELSE
    IF units <= 100 THEN
      energy = units * 2
    ELSE IF units <= 200 THEN
      energy = (100 * 2) + ((units - 100) * 3)
    ELSE
      energy = (100 * 2) + (100 * 3) + ((units - 200) * 5)
    END IF

    bill = energy + 50

    IF units > 300 THEN
      bill = bill * 1.05
    END IF

    PRINT "Energy charge: Rs.", ROUND(energy, 2)
    PRINT "Total bill: Rs.", ROUND(bill, 2)
  END IF
END
```

**7. Python Source Code**

Python

```
# Tiered Electricity Billing System

# Read electricity consumption
units = int(input("Enter units consumed: "))

# Validate inputs
if units < 0:
  print("Invalid units")
else:
  # Compute slab-based tiered energy charge
  if units <= 100:
    energy = units * 2
  elif units <= 200:
    energy = 100 * 2 + (units - 100) * 3
  else:
    energy = 100 * 2 + 100 * 3 + (units - 200) * 5

  # Add flat meter service charge
  bill = energy + 50

  # Levy 5% surcharge for high consumption (>300 units)
  if units > 300:
```

```
    bill = bill * 1.05

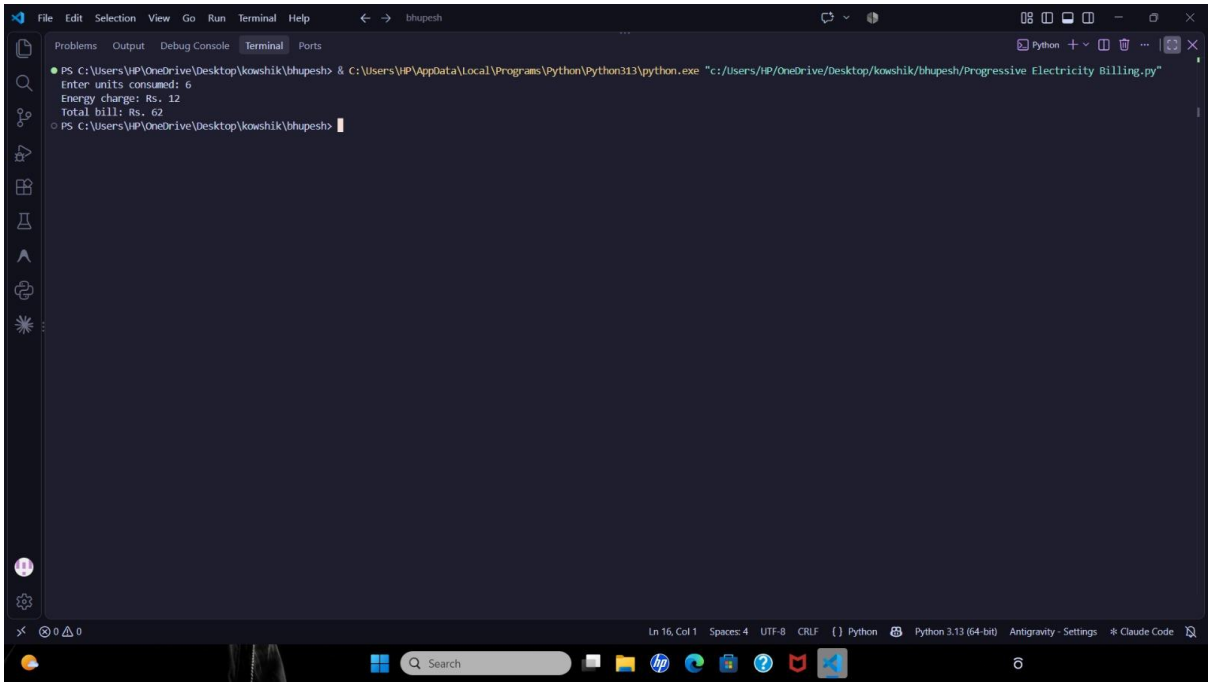
# Display billing breakdown
print("Energy charge: Rs.", round(energy, 2))
print("Total bill: Rs.", round(bill, 2))
```

**8. Test Cases**

| Test Case ID | Input Given | Expected Output | Actual Output | Status |
|---|---|---|---|---|
| **TC_01 (Slab 1)** | Units = 80 | Energy charge: Rs. 160<br><br>Total bill: Rs. 210 | Energy charge: Rs. 160<br><br>Total bill: Rs. 210 | Pass |
| **TC_02 (Slab 2)** | Units = 150 | Energy charge: Rs. 350<br><br>Total bill: Rs. 400 | Energy charge: Rs. 350<br><br>Total bill: Rs. 400 | Pass |

| Test Case ID | Input Given | Expected Output | Actual Output | Status |
|---|---|---|---|---|
| **TC_03 (Slab 3 - No Surcharge)** | Units = 250 | Energy charge: Rs. 750<br><br>Total bill: Rs. 800 | Energy charge: Rs. 750<br><br>Total bill: Rs. 800 | Pass |
| **TC_04 (Slab 3 - With Surcharge)** | Units = 350 | Energy charge: Rs. 1250<br><br>Total bill: Rs. 1365.0 | Energy charge: Rs. 1250<br><br>Total bill: Rs. 1365.0 | Pass |
| **TC_05 (Invalid Input)** | Units = -20 | Invalid units | Invalid units | Pass |

**9. Output Screenshots**




## 10. Conclusion

The Tiered Electricity Billing System was successfully implemented and verified. The script highlights the application of conditional statements, incremental arithmetic calculation for tiered structures, fixed surcharge adjustments, and high-load conditional penalties.